

ExamNoteX

ELECTRIC FIELD

Electric Field

Definition:

The region around a charge where another charge experiences electrostatic force is called the electric field.

Electric Field Intensity

Definition:

Electric field intensity at a point is the force experienced by unit positive charge placed at that point.

Formula:

$$E = F / q$$

SI Unit:

$$N/C$$

Direction of Electric Field

The direction of an electric field is the direction of force acting on positive test charge.

Electric Field Due to Point Charge

Formula:

$$E = kQ / r^2$$

Where:

- E = Electric field intensity
- Q = Source charge
- r = Distance from charge
- k = Electrostatic constant

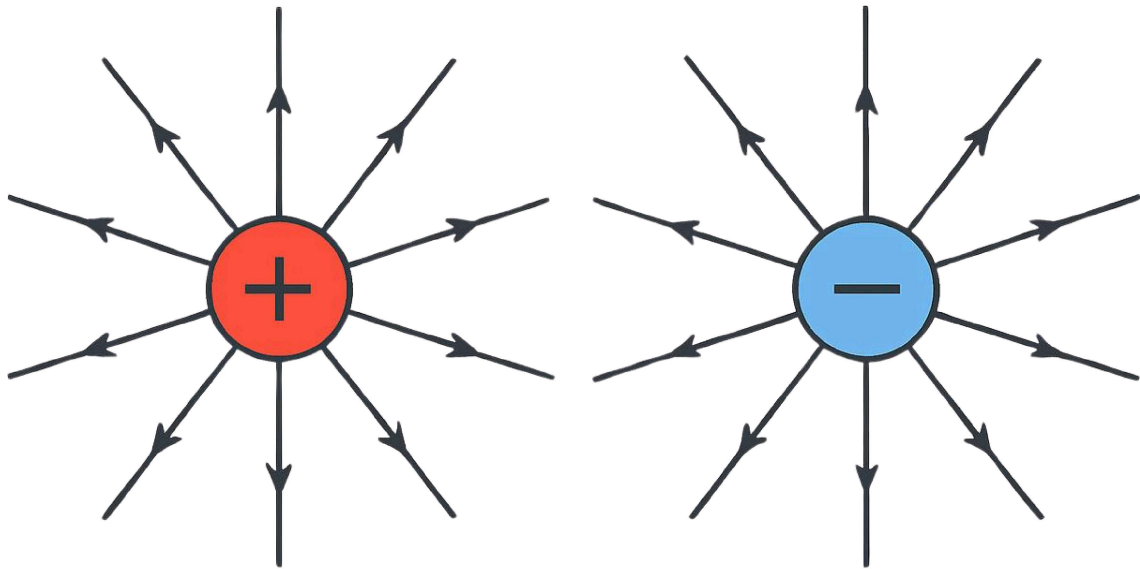
Electric Field Lines

Definition:

Imaginary lines representing the direction of an electric field are called electric field lines.

Properties:

- Field lines start from positive charge.
- Field lines end at negative charge.
- Field lines never intersect.
- Closer lines indicate stronger fields.



ELECTRIC FIELD

Electric Dipole

Definition:

A pair of equal and opposite charges separated by a small distance is called an electric dipole.

Dipole Moment:

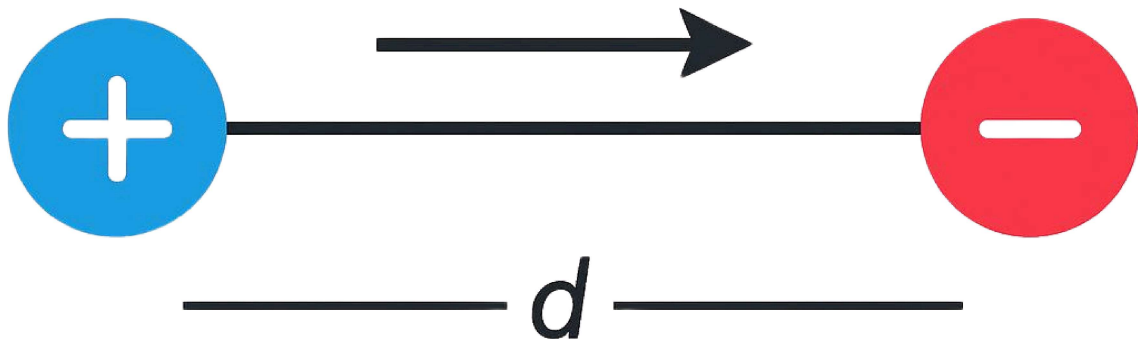
$$p = q \times 2l$$

Where:

- p = Dipole moment
- q = Magnitude of charge

- $2l$ = Distance between charges

dipole
moment



ELECTRIC DIPOLE

Electric Flux

Definition:

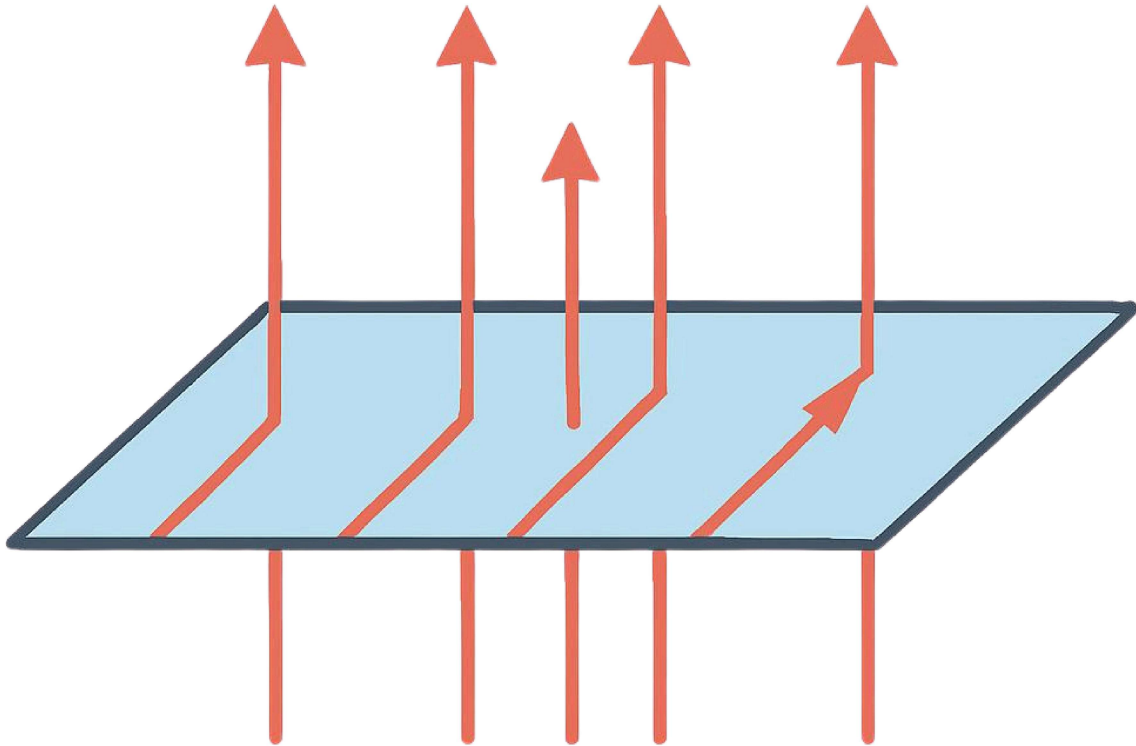
The total number of electric field lines passing through a surface is called electric flux.

Formula:

$$\Phi = E \cdot A$$

SI Unit:

$\text{N}\cdot\text{m}^2/\text{C}$



ELECTRIC FLUX

Applications of Electric Field

- Electrostatic painting
- Inkjet printers
- Electrostatic precipitators

Quick Revision Points

- An electric field exists around every charge.
- Electric field intensity is force per unit charge.
- Electric field due to point charge is inversely proportional to square of distance.
- Electric field lines never intersect.
- An electric dipole consists of equal and opposite charges.
- Dipole moment is represented by p .
- Electric flux measures field lines passing through the surface.

Important Formulae

- $E = F / q$
- $E = kQ / r^2$
- $p = q \times 2l$
- $\Phi = E \cdot A$

Conclusion

The electric field helps in understanding interaction between electric charges and forms the basis of electrostatics.